

UTI) $\rightarrow 23 \text{ BCS 116 N8}$

Exp-7

DSU
(gold)

První vs
Kruskal



- List <list <pair>> Adj.
- for (Height in flights)
 - adj.get(u).add(v, wt)
- use Priority Queue pq (MinHeap)
- pq.add(src, 0, 0)
 - node dist stops
- while priorityQueue (pq not empty)
 - cur = pq.poll()
 - for (nbr = adj[u])
 - if (stops < k) pq.add(nbr, dist + nbr.wt, cur.stops + 1)
 - if (nbr == dest)
 - return dist + nbr
- return -1

Time complexity = $O(E \log(V/R))$

1) for (arr)

while (arr.size > 1)

{ arr.sort()

return arr[0]}

for (i = 0; i < arr.size - 1; i++)

arr[i] = arr[i] + arr[i+1]}

arr = new ArrayList<>

return arr[0]}

O(N^2)